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TEMPERATURE MONITORING OF A VOLCANIC VENT GAS EMISSION USING TIME SERIES ANALYSIS AND MARKOV CHAIN APPROACH: AN EXAMPLE IN NISYROS VOLCANO, SOUTH AEGEAN

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Nisyros island is a volcanic island located in the South Aegean Sea in Greece. The island is located between Kos and Tilos and it is a part of Dodecanese group of islands in Greece. For monitoring the vent gas emission temperatures in Nisyros, a temperature sensor has been established in a foumarole site in the volcano of the island. For monitoring the volcanic temperature, hourly data have been used from the sensor measured from August of 2016 since January of 2017. Temperature time series were analyzed using two different methods to examine the main periodicities. One approach was a propabilistic method (Markov chain) and the second approach was time series decomposition (Kolmogorov-Zurbenko filter). Markov chain was used to prove probabilistically the periodicity in the temperature data and aggregate load impact in forecasting. With the time series decompose, we separate the long from the short cycle events using the Kolmogorov-Zurbenko filter. Both methodologies determined the main periodicities of the long cycle events. Using spectral analysis, we determine the main periodicities of the temperature data in the time series decomposition approach while using cluster analysis we determine the main periodicities in the Markov chain approach. From both methodologies, we have determined the main periodicities of the long term cycles of the temperature data are 8 days, and 12 days. Temperature measurements range between 31 ^oC and 62 ^oC. The 8-day periodicity shows a temperature range between 46 ^oC and 62 ^oC, while the 8-day periodicity reflects a range between 52 ^oC and 62 ^oC. Both periods are related with cyclones that occur in Aegean Sea. Since Nisyros is an island, the temperature measurement is probably affected by various phenomena (cyclones, sea tides) that occur in the sea. Using similar time series techniques for volcanic temperatures in different locations, we may reveal different periodicities depending on the location and landscape of the volcano.

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